

## Product Specification

Part Name: TFT LCD Display Module  
Customer Part ID:  
Zhongjingyuan Part ID: ZJY400-8532ACT  
Ver: A

|             |
|-------------|
| Customer:   |
| Approved by |

|                                                               |
|---------------------------------------------------------------|
| From: Zhengzhou Zhongjingyuan Electronic Technology Co., LTD. |
| Approved by                                                   |

**Zhengzhou Zhongjingyuan Electronic Technology Co., LTD.**

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Notes:

1. Please contact Zhengzhou Zhongjingyuan Electronic Technology Co., LTD. before assigning your product based on this module specification
2. The information contained herein is presented merely to indicate the characteristics and performance of our products. No responsibility is assumed by Zhengzhou Zhongjingyuan Electronic Technology Co., LTD. for any intellectual property claims or other problems that may result from application based on the module described herein.

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## Table of Contents

|                                                 |    |
|-------------------------------------------------|----|
| Revision History .....                          | 3  |
| 1 General Specifications .....                  | 4  |
| 2 Pin Assignment .....                          | 5  |
| 3 Absolute Maximum Ratings .....                | 6  |
| 4 Electrical Characteristics .....              | 6  |
| 5 INTERFACE TIMING .....                        | 9  |
| 5.1 System Bus Read/Write Characteristics. .... | 9  |
| 5.2 Power ON/OFF Timing .....                   | 10 |
| 6 Optical Characteristics .....                 | 12 |
| 7 Environmental / Reliability Test .....        | 16 |
| 8 Mechanical Drawing .....                      | 17 |
| 9 Precautions For Use of LCD Modules .....      | 18 |



## Revision History

| Rev | Issued Date   | Description          | Page | Editor |
|-----|---------------|----------------------|------|--------|
| 1.0 | Nov. 18, 2022 | First release        | All  |        |
| 2.0 | Apr. 18, 2024 | Change the thickness | 4/17 |        |
|     |               |                      |      |        |



## 1 General Specifications

| Feature                    |                                    | Specifications         |
|----------------------------|------------------------------------|------------------------|
| Display Spec.              | LCD type                           | 4.0 inch               |
|                            | Resolution (H*V)                   | 720(RGB) × 720         |
|                            | Technology Type                    | a-Si TFT               |
|                            | Pixel Configuration                | R.G.B. Vertical Stripe |
|                            | Display Mode                       | Normally Black         |
|                            | Viewing Direction                  | ALL                    |
|                            | Gray Scale Inversion Direction     | -                      |
| Mechanical Characteristics | OutlineDimensions (W x H x T) (mm) | 84*84*3.2              |
|                            | Active Area(mm)                    | 71.93*71.93            |
|                            | CTP View Area(mm)                  | \                      |
|                            | With /Without Touch screen         | With CTP               |
|                            | Connector Type (for LCD)           | 0.5mm pitch 40pin ZIF  |
|                            | Backlight Type                     | LED                    |
|                            | Weight (g)                         | TBD                    |
| Electrical Characteristics | Display Interface                  | RGB 18bits             |
|                            | Number of color                    | 262K                   |
|                            | Display Driver IC                  | NV3052CGRB             |
|                            | Touch Driver IC                    | FT6336U                |



## 2 Pin Assignment

|       |          |                                                                                                              |
|-------|----------|--------------------------------------------------------------------------------------------------------------|
| 1     | LED A    | LED ANODE                                                                                                    |
| 2     | LED K1   | LED CATHODE                                                                                                  |
| 3     | LED K2   | LED CATHODE                                                                                                  |
| 4     | GND      | Ground                                                                                                       |
| 5     | VCI      | Power supply (+3.3V)                                                                                         |
| 6     | RESET    | Reset Signal ,Active Low                                                                                     |
| 7     | IM1      | Not connection                                                                                               |
| 8     | IM0      | Not connection                                                                                               |
| 9     | SDA      | Not connection                                                                                               |
| 10    | SCK      | Not connection                                                                                               |
| 11    | CS       | Not connection                                                                                               |
| 12    | PCLK     | RGB dot clock signal                                                                                         |
| 13    | DE       | RGB data enable signal                                                                                       |
| 14    | VSYNC    | RGB frame synchronizing signal                                                                               |
| 15    | HSYNC    | RGB line synchronizing signal                                                                                |
| 16~33 | DB0~DB17 | RGB data signal (DB0: Blue LSB; DB5:Blue MSB; DB6: GREEN LSB; DB11: GREEN MSB; DB12: RED LSB; DB17: RED MSB) |
| 34    | GND      | Ground                                                                                                       |
| 35    | TP_INT   | Touch Interrupt                                                                                              |
| 36    | TP_SDA   | Touch IIC Data signal                                                                                        |
| 37    | TP_SCL   | Touch IIC Clock signal                                                                                       |
| 38    | TP_RESET | Touch Reset Signal                                                                                           |
| 39    | TP_VCI   | Touch Power supply                                                                                           |
| 40    | GND      | Ground                                                                                                       |



### 3 Absolute Maximum Ratings

GND=0V, Ta= 25°C

| Item                  | Symbol | Value     | Unit |
|-----------------------|--------|-----------|------|
| Operating temperature | Topr   | -20 to 70 | °C   |
| Storage temperature   | Tstg   | -30 to 80 | °C   |

**Note:** Note1: Absolute maximum rating is the limit value beyond which the IC maybe broken.

They do not assure operations.

Note2: Background color changes slightly depending on ambient temperature. This Phenomenon is reversible.

Ta ≤ 70°C: 75%RH max

Ta > 70°C: absolute humidity must be lower than the humidity of 75%RH at 70°C

Note3: Ta at -30°C will be <48hrs, at 80 °C will be <120hrs

### 4 Electrical Characteristics

#### 4.1 Driving TFT LCD Panel

| Item                  | Symbol          | Min                | Type | Max                | Unit | Test condition                                      |
|-----------------------|-----------------|--------------------|------|--------------------|------|-----------------------------------------------------|
| Operating voltage     | V <sub>DD</sub> | 2.5                | 3.3  | 3.6                | V    | -                                                   |
| Supply current        | I <sub>DD</sub> | -                  | 25.7 | 60                 | mA   | V <sub>DD</sub> =3.3V, Ta=25°C                      |
| Input voltage         | V <sub>IH</sub> | 0.8V <sub>DD</sub> | -    | V <sub>DD</sub>    | V    | -                                                   |
|                       | V <sub>IL</sub> | 0                  | -    | 0.2V <sub>DD</sub> | V    |                                                     |
| Input leakage current | I <sub>IL</sub> | -1.0               | -    | 1.0                | μA   | V <sub>IN</sub> =V <sub>DD</sub> or V <sub>SS</sub> |

Note: Voltage greater than above may damage the module.

All voltages are specified relative to VSS=0V.

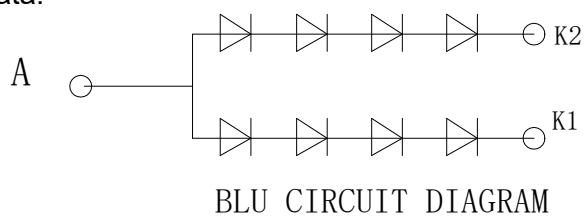


## 4.2 Driving Backlight

Ta=25℃

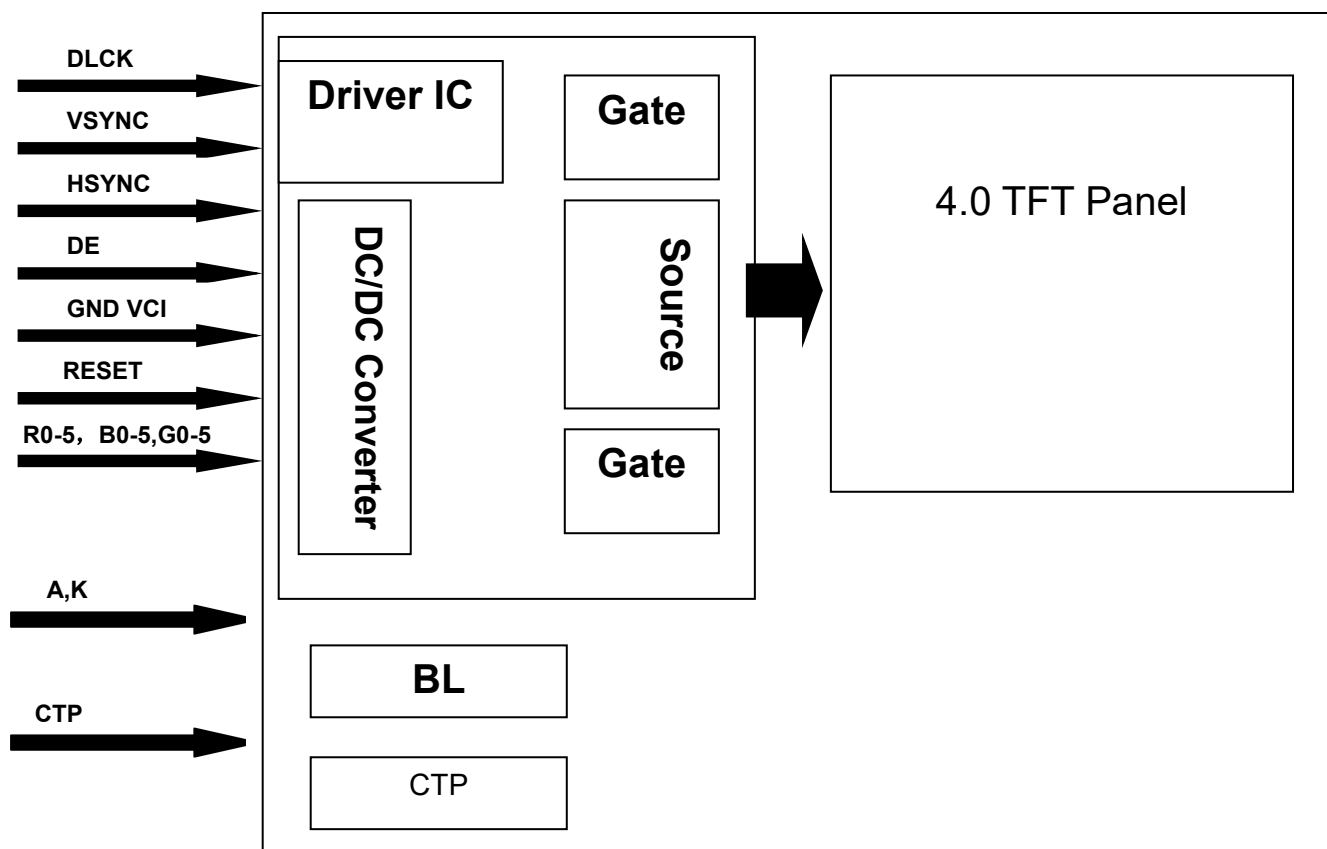
| Item            | Symbol | Min | Typ  | Max | Unit | Remark |
|-----------------|--------|-----|------|-----|------|--------|
| Forward Current | $I_F$  | --  | 40   | 50  | mA   |        |
| Forward Voltage | $V_F$  | -   | 12.8 | -   | V    |        |
| Connection mode |        | --  | 4S2P | --  |      |        |
| LED number      | /      |     | 8    |     | pcs  |        |

Note1: Optical performance should be evaluated at Ta=25℃ only .If LED is driven by high current, high ambient temperature & humidity condition. The life time of LED will be reduced. Operating life means brightness goes down to 50% initial brightness. Typical operating life time is estimated data.





### 4.3 Block Diagram



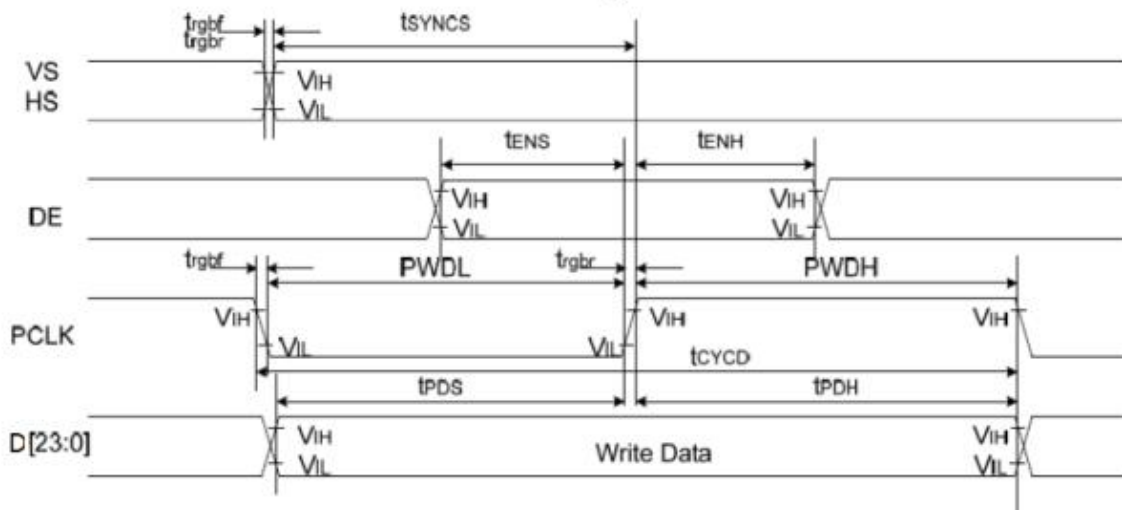




## 5 INTERFACE TIMING

### 5.1 System Bus Read/Write Characteristics.

Parallel 24/18/16-bit RGB Interface Timing Characteristics



| Signal  | Symbol      | Parameter                 | min | max | Unit | Description                         |
|---------|-------------|---------------------------|-----|-----|------|-------------------------------------|
| VS/HS   | tSYNCS      | VS/HS setup time          | 5   | -   | ns   | 24/18/16-bit bus RGB interface mode |
|         | tSYNCH      | VS/HS hold time           | 5   | -   | ns   |                                     |
| DE      | tENS        | DE setup time             | 5   | -   | ns   |                                     |
|         | tENH        | DE hold time              | 5   | -   | ns   |                                     |
| D[23:0] | tPOS        | Data setup time           | 5   | -   | ns   |                                     |
|         | tpDH        | Data hold time            | 5   | -   | ns   |                                     |
| PCLK    | PWDH        | PCLK high-level period    | 13  | -   | ns   |                                     |
|         | PWDL        | PCLK low-level period     | 13  | -   | ns   |                                     |
|         | tCYCD       | PCLK cycle time           | 28  | -   | ns   |                                     |
|         | tgrbr,tgrbf | PCLK,HS,VS rise/fall time | -   | 15  | ns   |                                     |

Note 1: IOVCC=1.65 to 3.6V, VCI=2.5 to 6V, VSSA=VSS=0V, Ta=-30 to 70℃

## 5.2 Power On/Off Sequence

IOVCC and VCI can be applied in any order. IOVCC and VCI can be powered down in any order.

During power off, if LCD is in the Sleep Out mode, VCI and IOVCC must be powered down minimum 120msec after RESX has been released.

During power off, if LCD is in the Sleep In mode, IOVCC or VCI can be powered down minimum 0msec after RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

Note 1: There will be no damage to the display module if the power sequences are not met.

Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command.

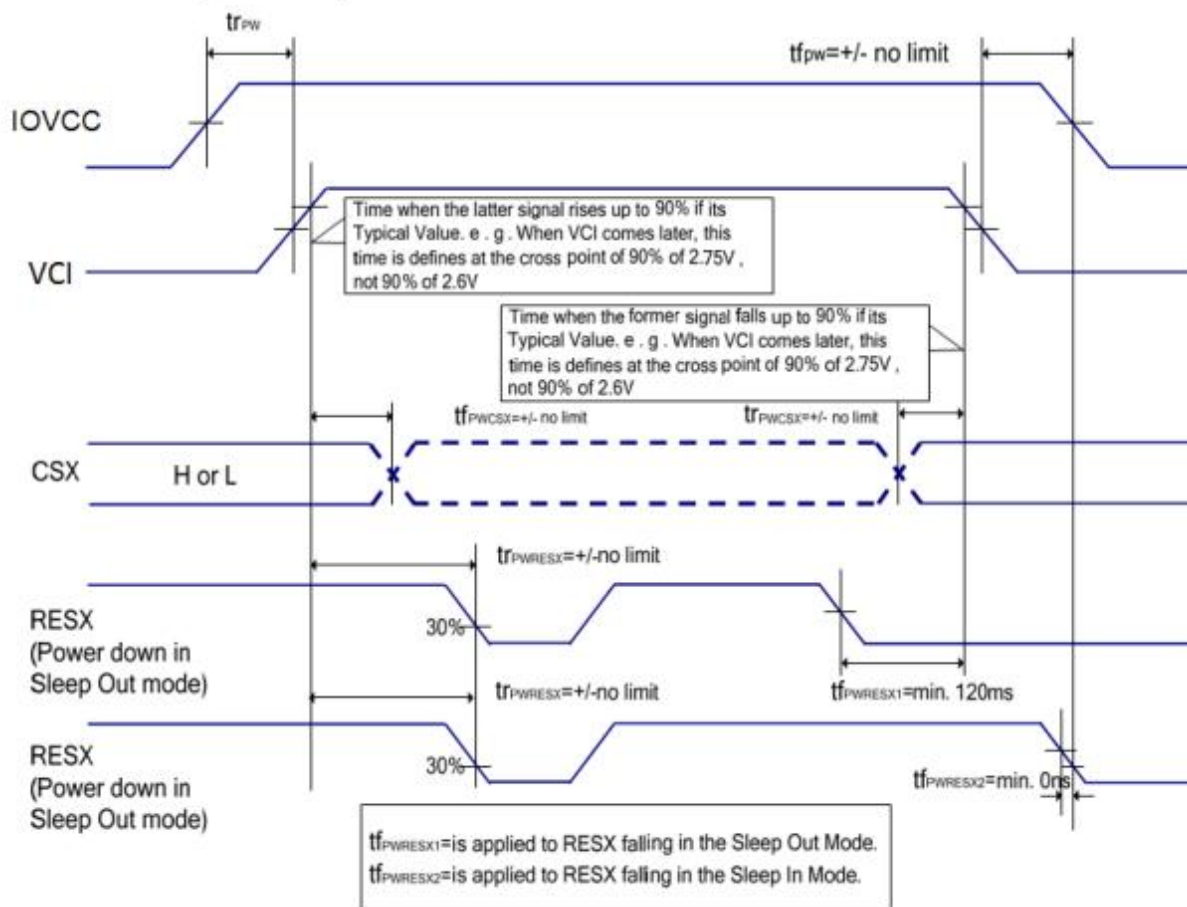
Also between receiving Sleep In command and Power Off Sequence.

If RESX line is not held stable by host during Power On Sequence, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below:

### Case 1 – RESX line is held high or unstable by host at power on

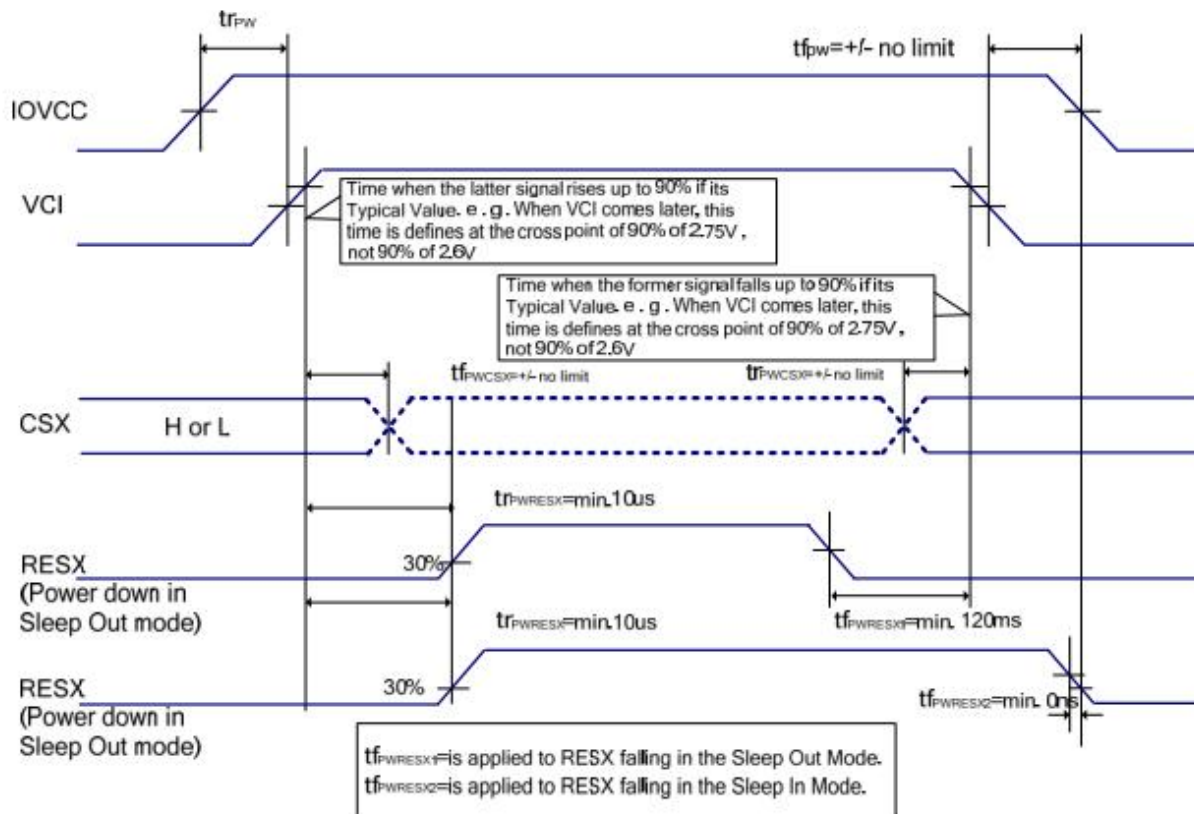
If RESX line is held High or unstable by the host during Power On, then a Hardware Reset must be applied after both VCI and IOVCC have been applied – otherwise correct functionality is not guaranteed. There is no timing restriction upon this hardware reset.





## Case 2 – RESX line is held low or unstable by host at power on

If RESX line is held Low (and stable) by the host during Power On, then the RESX must be held low for minimum 10sec after both VCI and IOVCC have been applied.



## Uncontrolled power off

The uncontrolled power off means a situation when e.g. there is removed a battery without the controlled power off sequence. There will not be any damages for the display module or the display module will not cause any damages for the host or lines of the interface.

At an uncontrolled power off the display will go blank and there will not be any visible effects within (TBD) second on the display (blank display) and remains blank until "Power On Sequence" powers it up.



## 6 Optical Characteristics

Ta=25℃

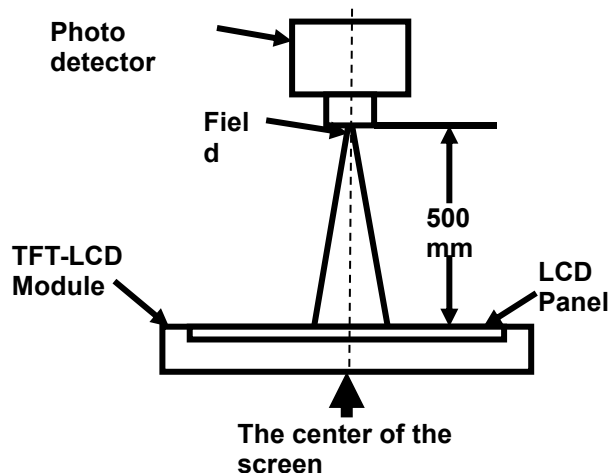
| Item           | Symbol     | Condition        | Min | Typ | Max | Unit              | Remark         |
|----------------|------------|------------------|-----|-----|-----|-------------------|----------------|
| View Angles    | $\theta T$ | $CR \geq 10$     | -   | 80  | -   | Degree            | Note 2         |
|                | $\theta B$ |                  | -   | 80  | -   |                   |                |
|                | $\theta L$ |                  | -   | 80  | -   |                   |                |
|                | $\theta R$ |                  | -   | 80  | -   |                   |                |
| Contrast Ratio | CR         | $\theta=0^\circ$ | 700 | 900 | -   | -                 | Note1<br>Note3 |
| Response Time  | $T_{ON}$   | 25℃              | -   | 25  | 35  | ms                | Note1          |
|                | $T_{OFF}$  |                  |     |     |     |                   | Note4          |
| Uniformity     | U          | -                | 70  | 80  | -   | %                 | Note1<br>Note6 |
| NTSC           | -          | -                | -   | 68  | -   | %                 | Note 5         |
| Luminance      | L          |                  | 250 | -   | -   | cd/m <sup>2</sup> | Note1<br>Note7 |

Test Conditions:

1.  $V_F=12.8V$ ,  $I_F=40mA$ , the ambient temperature is 25℃.
2. The test systems refer to Note 1 and Note 2.

**Note 1: Definition of optical measurement system.**

The optical characteristics should be measured in dark room. After 5 minutes operation, the optical properties are measured at the center point of the LCD screen. All input terminals LCD panel must be ground when measuring the center area of the panel.



| Item           | Photo detector | Field |
|----------------|----------------|-------|
| Contrast Ratio | SR-3A          | 1°    |
| Luminance      |                |       |
| Chromaticity   |                |       |
| Lum Uniformity |                |       |
| Response Time  | BM-7A          | 2°    |

**Note 2: Definition of viewing angle range and measurement system.**

viewing angle is measured at the center point of the LCD by CONOSCOPE(ergo-80).

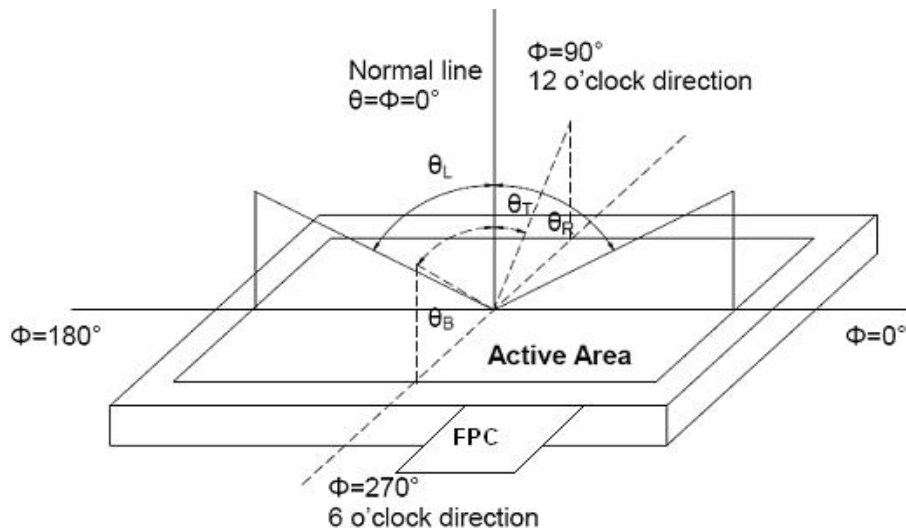


Fig. 1 Definition of viewing angle

**Note 3: Definition of contrast ratio**

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD is on the "White" state}}{\text{Luminance measured when LCD is on the "Black" state}}$$

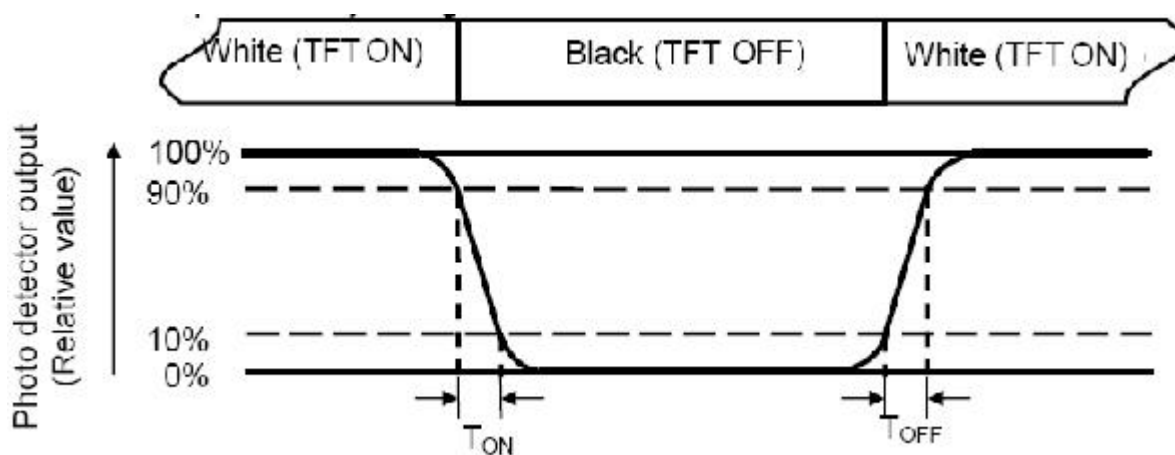
"White state ":The state is that the LCD should be driven by Vwhite.

"Black state": The state is that the LCD should be driven by Vblack.

Vwhite: To be determined    Vblack: To be determined.

**Note 4: Definition of Response time**

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (TON) is the time between photo detector output intensity changed from 90% to 10%. And fall time (TOFF) is the time between photo detector output intensity changed from 10% to 90%.

**Note 5: Definition of color chromaticity (CIE1931)**

Color coordinates measured at center point of LCD.

**Note 6: Definition of Luminance Uniformity**

Active area is divided into 9 measuring areas (Refer Fig. 2). Every measuring point is placed at the center of each measuring area.

Luminance Uniformity (U) =  $L_{\min} / L_{\max}$

L-----Active area length W----- Active area width

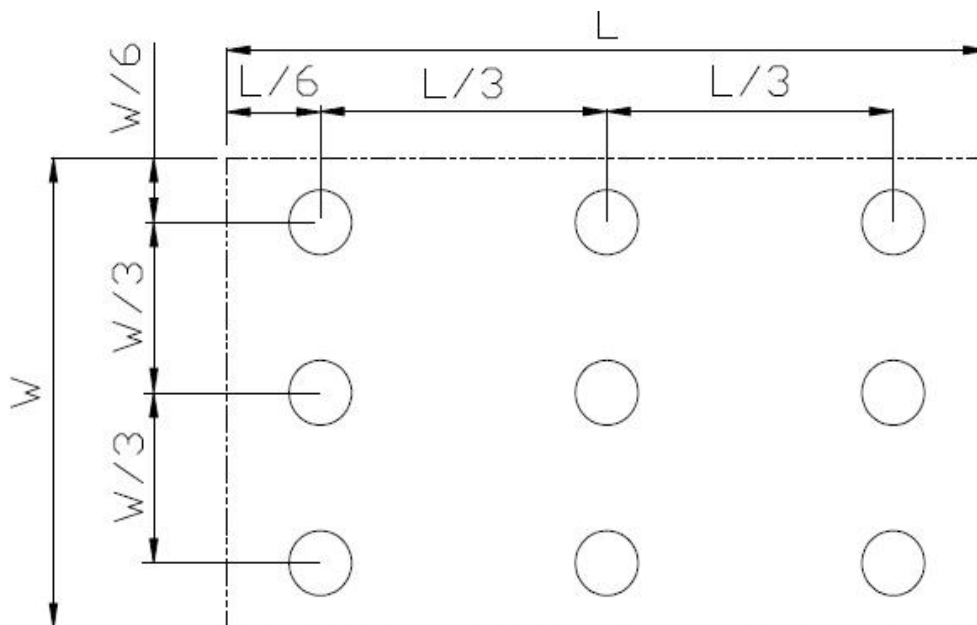


Fig. 2 Definition of uniformity

$L_{\max}$ : The measured maximum luminance of all measurement position.

$L_{\min}$ : The measured minimum luminance of all measurement position.

**Note 7: Definition of Luminance:**

Measure the luminance of white state at center point.





## 7 Environmental / Reliability Test

| Item                            | Condition                                            | Time (hrs) | Assessment                                         |
|---------------------------------|------------------------------------------------------|------------|----------------------------------------------------|
| High temp. Storage              | 80°C                                                 | 120        | No abnormalities<br>in functions<br>and appearance |
| High temp. Operating            | 70°C                                                 | 120        |                                                    |
| Low temp. Storage               | -30°C                                                | 120        |                                                    |
| Low temp. Operating             | -20°C                                                | 120        |                                                    |
| Humidity                        | 40°C/ 90%RH                                          | 120        |                                                    |
| Thermal<br>Shock(Non-operation) | -20°C ← 25°C → 70°C<br>(0.5 hour ← 5 min → 0.5 hour) | 10cycles   |                                                    |

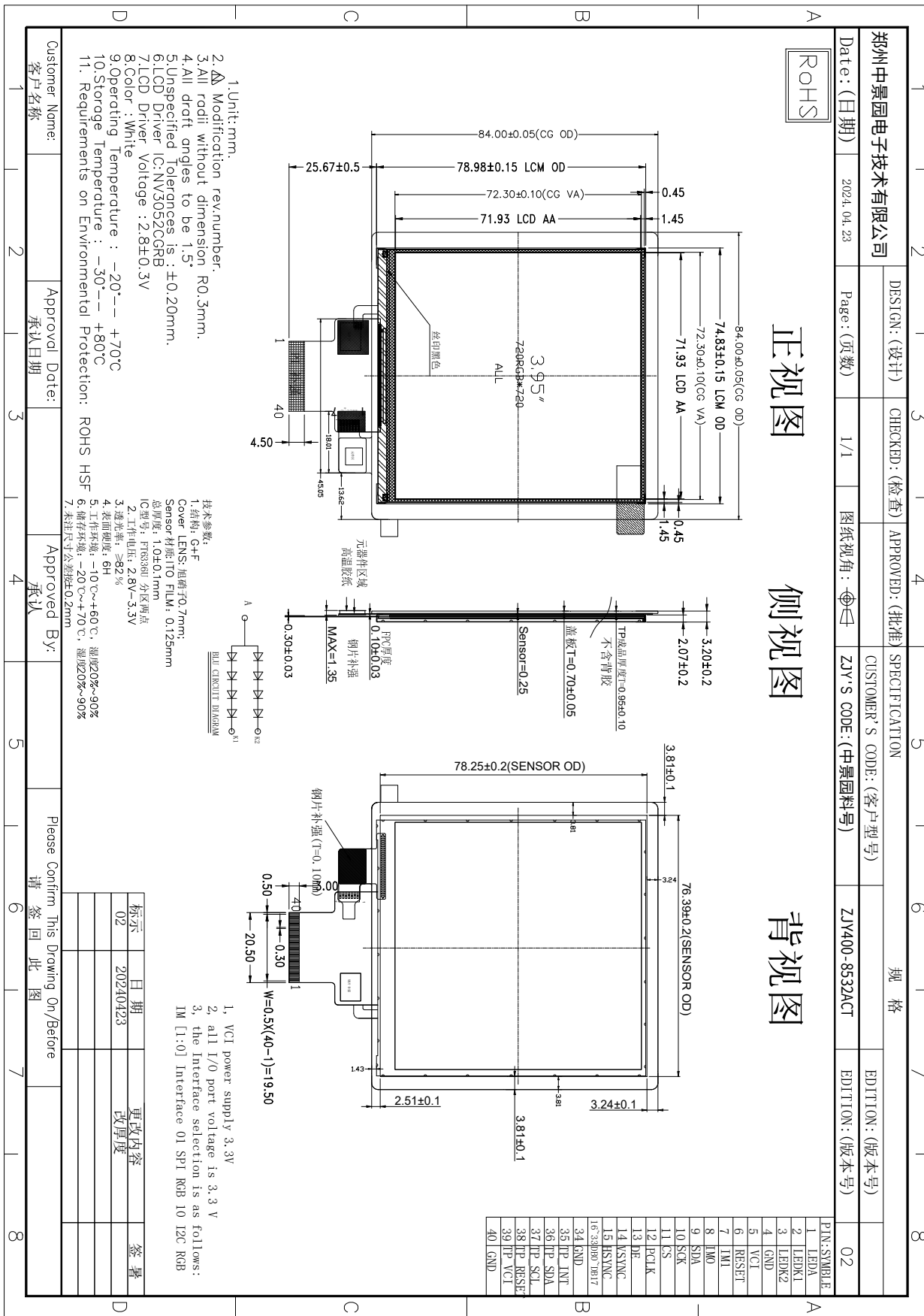
Remark:

- 1.The test samples should be applied to only one test item.
- 2.Sample size for each test item is 1~10pcs.
- 3.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.





## 8 Mechanical Drawing





## 9 Precautions For Use of LCD Modules

### 9.1 Handling Precautions

9.1.1.1 The display panel is made of glass. Do not subject it to a mechanical shock by dropping it from a high place, etc.

9.1.1.2 If the display panel is damaged and the liquid crystal substance inside it leaks out, be sure not to get any in your mouth, if the substance comes into contact with your skin or clothes, promptly wash it off using soap and water.

9.1.1.3 Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary.

9.1.1.4 The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully.

9.1.1.5 If the display surface is contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If still not completely clear, moisten cloth with one of the following solvents:

- Isopropyl alcohol
- Ethyl alcohol

Solvents other than those mentioned above may damage the polarizer. Especially, do not use the following:

- Water
- Ketone
- Aromatic solvents

9.1.1.6 Do not attempt to disassemble the LCD Module.

9.1.1.7 If the logic circuit power is off, do not apply the input signals.

9.1.1.8 To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment.

9.1.1.9 Be sure to ground the body when handling the LCD Modules.

9.1.1.10 Tools required for assembly, such as soldering irons, must be properly ground.

9.1.1.11 To reduce the amount of static electricity generated, do not conduct assembly and other work under dry conditions.

9.1.1.12 The LCD Module is coated with a film to protect the display surface. Be care when peeling off this protective film since static electricity may be generated.

9.1.1.13 Storage precautions

9.1.1.14 When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps.

9.1.1.15 The LCD modules should be stored under the storage temperature range. If the LCD modules will be stored for a long time, the recommend condition is:

9.1.1.16 The LCD modules should be stored in the room without acid, alkali and harmful gas.

### 9.2 Transportation Precautions

The LCD modules should be no falling and violent shocking during transportation, and also should avoid excessive press, water, damp and sunshine.